

Lab 19: Halving Algorithm

Aim

To implement the **Halving Algorithm** for finding a consistent hypothesis from the given training examples.

Algorithm

1. Define the training examples and their classifications.
2. Initialize the hypothesis using the first positive example.
3. Compare each positive example with the current hypothesis.
4. If an attribute differs, replace it with ?.
5. Continue until all positive examples are processed.
6. Display the final consistent hypothesis.

Program

```
# Lab 19: Halving Algorithm
```

```
import numpy as np
```

```
# Training data
```

```
data = np.array([
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
])
```

```
# Initialize hypothesis
```

```
hypothesis = ['0'] * 6
```

```
# Process positive examples
```

```
for row in data:
```

```
    if row[-1] == 'Yes':
```

```
        for i in range(6):
```

```
            if hypothesis[i] == '0':
```

```
                hypothesis[i] = row[i]
```

```
            elif hypothesis[i] != row[i]:
```

```
                hypothesis[i] = '?'
```

```
print("Final Hypothesis:", hypothesis)
```

Sample Output

```
Final Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

Result

Thus, the **Halving Algorithm was successfully implemented**, and a consistent hypothesis was obtained from the given training examples.